Lecture 19: Normal Matrices

Hermitian Geometry and Spectral Projections

Today's Question

From previous lectures, spectral decomposition tries to split a matrix into pieces:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k.$$

Then every vector splits into eigenspace components:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

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Are these pieces perpendicular?

Or are they only oblique algebraic pieces?

Why Study Normal Matrices?

Diagonalizable matrices already give spectral decomposition.

But diagonalization alone does **not** promise geometry:

$$\mathbb{C}^n = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k} \qquad \text{may be oblique.}$$

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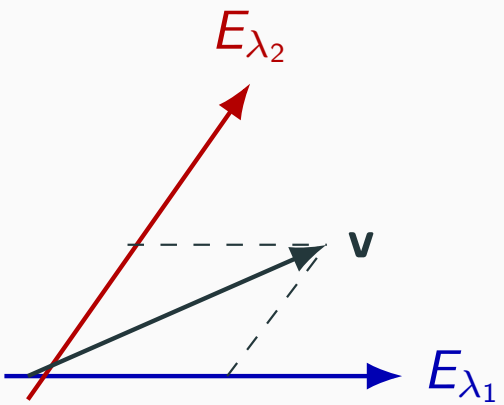
Normal matrices are the matrices where spectral decomposition becomes **orthogonal geometry**:

$$\boxed{\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.}$$

This is why normal matrices matter.

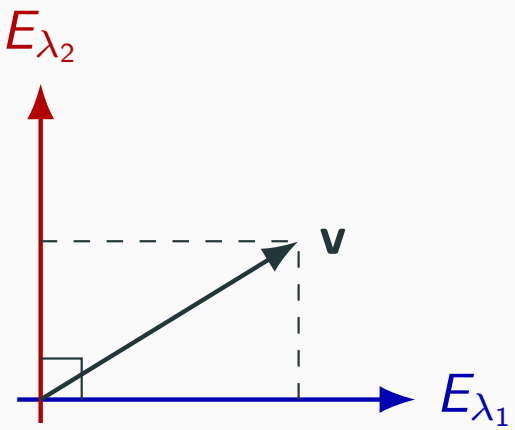
Two Pictures to Keep in Mind

Oblique eigenspaces



algebra works
geometry tilted

Orthogonal eigenspaces



clean geometry
perpendicular pieces

Normal matrices answer: when does the right picture happen?

The Surprising Theorem

Main Theorem We Will Prove

For a complex matrix A,

$$A \text{ normal} \quad \Longleftrightarrow \quad \text{all spectral projections } P_i \text{ satisfy } P_i^H = P_i.$$

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Equivalently:

$$A \text{ normal} \quad \Longleftrightarrow \quad \text{eigenspaces of } A \text{ are mutually Hermitian-orthogonal.}$$

Not just diagonalizable: diagonalizable with orthogonal spectral pieces.

How This Connects to Matrix Exponentials

Lecture 18 gave the spectral formula:

$$e^{As} = e^{\lambda_1 s} P_1 + \cdots + e^{\lambda_k s} P_k.$$

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If the projections are oblique, the algebra works but the geometry is distorted.

If A is normal, then each P_i is an orthogonal projection:

$$P_i^H = P_i.$$

Meaning: the system evolves along perpendicular eigenspace directions.

Complex Length

Why Are We in the Complex World?

Our spectral method needs roots of the characteristic polynomial:

$$\det(tI - A) = 0.$$

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$$t^2 + 1 = 0 \qquad \text{has no real root.}$$

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Over $\mathbb{C}$, every polynomial factors into linear factors.

So the spectral story naturally lives over $\mathbb{C}$.

But Complex Geometry Needs Repair

We want to measure:

- **length**: how big is a vector?
- **angle**: when are two vectors perpendicular?

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Over $\mathbb{C}$, this formula breaks length.

Hermitian geometry is the repaired geometry of complex vector spaces.

The Old Length Formula

Over real numbers, length comes from

$$\|\mathbf{v}\|^2 = \mathbf{v}^T\mathbf{v}.$$

For

$$\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n,$$

we get

$$\mathbf{v}^T\mathbf{v} = x_1^2 + x_2^2 + \cdots + x_n^2.$$

Key: a nonzero real vector has positive length squared.

What Goes Wrong over $\mathbb{C}$?

Try the same formula on a complex vector:

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$$\mathbf{v}^T\mathbf{v} = \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = 1 + i^2.$$

But $i^2 = -1$, so

$$\mathbf{v}^T\mathbf{v} = 1 - 1 = 0.$$

Problem: a nonzero vector has zero length.

The Missing Move

The problem is this multiplication:

$$i \cdot i = i^2 = -1.$$

For length, we want a positive number:

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$$i \quad \longmapsto \quad -i.$$

Then

$$(-i) \cdot i = 1.$$

Key: complex length needs conjugation before dot product.

Hermitian Transpose of a Vector

Definition: Hermitian Transpose of a Vector

For a complex column vector $\mathbf{v}$, define

$$\mathbf{v}^H := \overline{\mathbf{v}}^T.$$

Immediate example:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix} \qquad \Longrightarrow \qquad \mathbf{v}^H = \begin{pmatrix} 1 & -i \end{pmatrix}.$$

The letter H means **Hermitian**.

The Same Vector Now Has Positive Length

For

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \mathbf{v}^H = \begin{pmatrix} 1 & -i \end{pmatrix}.$$

Now compute:

$$\mathbf{v}^H \mathbf{v} = \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = 1 + (-i)i = 2.$$

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No zero-length nonzero vector.

Key: H repairs geometry over $\mathbb{C}$.

Hermitian Inner Product

Definition: Hermitian Inner Product

For complex vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$, define

$$\langle \mathbf{v}, \mathbf{w} \rangle_H := \mathbf{v}^H \mathbf{w}.$$

If

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

then

$$\langle \mathbf{v}, \mathbf{v} \rangle_H = \mathbf{v}^H \mathbf{v} = |v_1|^2 + \cdots + |v_n|^2.$$

Key: this is always positive when $\mathbf{v} \neq \mathbf{0}$.

Hermitian Orthogonal Vectors

Definition: Hermitian Orthogonal

Two complex vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ are **Hermitian orthogonal** if

$$\mathbf{v}^H \mathbf{w} = 0.$$

Example:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

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Then

$$\mathbf{v}^H \mathbf{w} = \begin{pmatrix} 1 & -i \end{pmatrix} \begin{pmatrix} 1 \\ -i \end{pmatrix} = 1 + (-i)(-i) = 1 + i^2 = 0.$$

Hermitian Orthogonal Subspaces

Definition: Hermitian Orthogonal Subspaces

Two subspaces $V, W \subseteq \mathbb{C}^n$ are **Hermitian orthogonal** if

$$\mathbf{v}^H \mathbf{w} = 0$$

for every $\mathbf{v} \in V$ and every $\mathbf{w} \in W$.

Notation:

$$V \perp_H W.$$

This is the geometry we want for eigenspaces.

Hermitian Conjugate

Hermitian Conjugate of a Matrix

Definition: Hermitian Conjugate

For a complex matrix A, define

$$A^H := \overline{A^T}.$$

This means:

transpose first, then **conjugate every entry**.

This is also called the **conjugate transpose**.

Concrete Example

Let

$$A = \begin{pmatrix} 1 & i \\ 2-i & 3 \end{pmatrix}.$$

First transpose:

$$A^T = \begin{pmatrix} 1 & 2-i \\ i & 3 \end{pmatrix}.$$

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First transpose:

$$A^T = \begin{pmatrix} 1 & 2-i \\ i & 3 \end{pmatrix}.$$

Then conjugate:

$$A^H = \overline{A^T} = \begin{pmatrix} 1 & 2+i \\ -i & 3 \end{pmatrix}.$$

Real Matrices Are the Old Case

If all entries of A are real, conjugation changes nothing.

So

$$A^H = \overline{A^T} = A^T.$$

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Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad \Longrightarrow \qquad A^H = A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

Key: Hermitian conjugate is the complex version of transpose.

The Product Rule Reverses Order

For matrices,

$$(AB)^H = B^H A^H.$$

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Why order reverses:

$$(AB)^H = \overline{(AB)^T} = \overline{B^T A^T} = \overline{B^T}\,\overline{A^T} = B^H A^H.$$

Key: Hermitian conjugate reverses multiplication order.

Why A^H Is the Correct Mirror

Hermitian conjugate is the operation that moves A across the inner product:

$$\langle A\mathbf{v}, \mathbf{w} \rangle_H = (A\mathbf{v})^H \mathbf{w}.$$

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$$(A\mathbf{v})^H \mathbf{w} = \mathbf{v}^H A^H \mathbf{w}.$$

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Use the product rule:

$$(A\mathbf{v})^H \mathbf{w} = \mathbf{v}^H A^H \mathbf{w}.$$

So

$$\langle A\mathbf{v}, \mathbf{w}\rangle_H = \langle \mathbf{v}, A^H \mathbf{w}\rangle_H.$$

Key: A^H is the mirror-action of A.

Normal Matrices

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Definition: Normal Matrix

A complex square matrix A is **normal** if

$$AA^H = A^H A.$$

In words:

A commutes with its Hermitian conjugate.

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This is not the final theorem.

The real question: what does this condition do to spectral projections?

First Special Case: Hermitian Symmetric

Definition: Hermitian Symmetric

A matrix A is **Hermitian symmetric** if

$$A^H = A.$$

Example:

$$A = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix}, \qquad A^H = A.$$

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Then A is automatically normal:

$$AA^H = A^2 = A^H A.$$

Second Special Case: Skew-Hermitian Symmetric

Definition: Skew-Hermitian Symmetric

A matrix A is **skew-Hermitian symmetric** if

$$A^H = -A.$$

Example:

$$A = \begin{pmatrix} i & 2+i \\ -2+i & 3i \end{pmatrix}, \qquad A^H = -A.$$

Second Special Case: Skew-Hermitian Symmetric

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Then A is automatically normal:

$$AA^H = A(-A) = -A^2 = (-A)A = A^H A.$$

Third Special Case: Unitary

Definition: Unitary Matrix

A matrix U is **unitary** if

$$U^H U = U U^H = I.$$

Example:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Third Special Case: Unitary

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Example:

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Unitary matrices are automatically normal:

$$UU^H = I = U^H U.$$

A Matrix That Is Not Normal

Let

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since J is real,

$$J^H = J^T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

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Since J is real,

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Compute both orders:

$$JJ^H = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad J^H J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

So $JJ^H \neq J^H J$. This matrix is **not normal**.

Diagonalizability

Before Spectral Projections

We want to use spectral projections:

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We want to use spectral projections:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

But this only makes sense after we know:

$$A \text{ is diagonalizable.}$$

So first we prove:

Theorem

Every normal matrix over $\mathbb{C}$ is diagonalizable.

Method: characteristic polynomial $\to$ radical polynomial $\to$ nilpotent normal matrix.

Positivity Lemma for Matrices

Positivity Lemma

For any complex matrix M,

$$MM^H = 0 \quad \Longrightarrow \quad M = 0.$$

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$$MM^H = 0 \quad \Longrightarrow \quad M = 0.$$

Why? The i-th diagonal entry of MM^H is

$$\mathrm{row}_i(M)\,\mathrm{row}_i(M)^H = \|\mathrm{row}_i(M)\|_H^2.$$

If $MM^H = 0$, every row has Hermitian length 0.

Therefore every row is zero.

Nilpotent Matrices

Definition: Nilpotent Matrix

A matrix N is **nilpotent** if

$$N^m = 0$$

for some positive integer m.

Example:

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

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Example:

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

Nilpotent means: repeat the action enough times, and everything disappears.

Polynomials Preserve Normality

Proposition

If A is normal, then every polynomial $f(A)$ is normal.

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Reason:

$$AA^H = A^H A \quad \Longrightarrow \quad A^m (A^H)^n = (A^H)^n A^m.$$

Also

$$f(A)^H = \overline{f}(A^H).$$

So $f(A)$ commutes with $f(A)^H$.

Normal Nilpotent Matrices Are Zero

Lemma

If N is normal and nilpotent, then

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Choose a power of two larger than m:

$$2^r \geq m.$$

Normal Nilpotent Matrices Are Zero

Lemma

If N is normal and nilpotent, then

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Choose a power of two larger than m:

$$2^r \geq m.$$

Then

$$N^{2^r} = 0.$$

Goal: descend from $N^{2^r} = 0$ to $N = 0$.

Step 1: Make a Positive Product

Since N is normal,

$$NN^H = N^H N.$$

So powers of N commute with powers of N^H.

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From $N^{2^r} = 0$, multiply by $(N^H)^{2^r}$:

$$N^{2^r}(N^H)^{2^r} = 0.$$

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So powers of N commute with powers of N^H.

From $N^{2^r} = 0$, multiply by $(N^H)^{2^r}$:

$$N^{2^r}(N^H)^{2^r} = 0.$$

Because the factors commute, this is

$$\left(N^{2^{r-1}}(N^H)^{2^{r-1}}\right)\left(N^{2^{r-1}}(N^H)^{2^{r-1}}\right) = 0.$$

Step 2: Recognize MM^H

Set

$$M = N^{2^{r-1}}.$$

Then

$$M^H = (N^H)^{2^{r-1}}.$$

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Set

$$M = N^{2^{r-1}}.$$

Then

$$M^H = (N^H)^{2^{r-1}}.$$

So the previous equation becomes

$$(MM^H)(MM^H) = 0.$$

Since $(MM^H)^H = MM^H$, this is

$$(MM^H)(MM^H)^H = 0.$$

Step 3: Positivity Gives One Descent

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Apply positivity again:

$$MM^H = 0 \quad \Longrightarrow \quad M = 0.$$

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$$(MM^H)(MM^H)^H = 0,$$

positivity gives

$$MM^H = 0.$$

Apply positivity again:

$$MM^H = 0 \quad \Longrightarrow \quad M = 0.$$

But $M = N^{2^{r-1}}$, so

$$N^{2^r} = 0 \quad \Longrightarrow \quad N^{2^{r-1}} = 0.$$

One power of two has been removed.

Induction: Repeat the Descent

The same argument works at every level:

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Apply it repeatedly:

$$N^{2^r} = 0 \Longrightarrow N^{2^{r-1}} = 0 \Longrightarrow \cdots \Longrightarrow N^2 = 0 \Longrightarrow N = 0.$$

Conclusion

Normal nilpotent matrices must be zero.

The Radical Polynomial

Write the characteristic polynomial honestly as a determinant:

$$\det(tI - A) = \prod_{i=1}^{k} (t - \lambda_i)^{m_i}.$$

The **radical polynomial** removes repeated powers:

$$r_A(t) = \prod_{i=1}^{k} (t - \lambda_i).$$

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Previous lecture criterion:

Diagonalizability Criterion

$$A \text{ diagonalizable} \quad \Longleftrightarrow \quad r_A(A) = 0.$$

Normal Implies Diagonalizable

Set

$$N = r_A(A) = \prod_{i=1}^{k}(A - \lambda_i I).$$

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By Cayley–Hamilton,

$$\det(tI - A)|_{t=A} = 0.$$

So some power of N is zero:

$$N^m = 0.$$

Normal Implies Diagonalizable

Set

$$N = r_A(A) = \prod_{i=1}^{k}(A - \lambda_i I).$$

By Cayley–Hamilton,

$$\det(tI - A)|_{t=A} = 0.$$

So some power of N is zero:

$$N^m = 0.$$

Also N is a polynomial in normal A, so N is normal.

Normal + nilpotent $\Rightarrow N = 0$.

Now Spectral Decomposition Is Legitimate

We proved:

$$A \text{ normal} \quad \Longrightarrow \quad r_A(A) = 0.$$

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We proved:

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Therefore A is diagonalizable.

Now we may use spectral projections:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k, \qquad P_i = f_i(A).$$

Next question: are these projections oblique or orthogonal?

Spectral Projections

Use Projections, Not Eigenvector Chasing

Now A is diagonalizable, so previous lectures give:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k.$$

Each P_i is the spectral projection onto the λ_i-eigenspace:

$$\mathrm{Im}(P_i) = E_{\lambda_i}.$$

Use Projections, Not Eigenvector Chasing

Now A is diagonalizable, so previous lectures give:

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Each P_i is the spectral projection onto the λ_i-eigenspace:

$$\mathrm{Im}(P_i) = E_{\lambda_i}.$$

And each projection is a polynomial in A:

$$P_i = f_i(A).$$

This is the route: study the projections, not individual eigenvectors.

What the Projections Do

The projections split every vector into eigenspace pieces:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

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The projections split every vector into eigenspace pieces:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

They satisfy:

$$P_i^2 = P_i, \qquad P_iP_j = 0 \quad (i \neq j), \qquad P_1 + \cdots + P_k = I.$$

What the Projections Do

The projections split every vector into eigenspace pieces:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

They satisfy:

$$P_i^2 = P_i, \qquad P_iP_j = 0 \quad (i \neq j), \qquad P_1 + \cdots + P_k = I.$$

The question is:

Are these projections oblique, or orthogonal?

Normal matrices force the orthogonal case.

Polynomial Normality Applies Again

We already proved:

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Spectral projections are exactly such polynomials:

$$P_i = f_i(A).$$

So normality passes from A to every P_i.

Apply This to Spectral Projections

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If A is normal, then by the proposition:

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But P_i is also a projection:

$$P_i^2 = P_i.$$

Now we need one lemma: normal projection $\Rightarrow$ Hermitian projection.

Normal Projection Lemma

Lemma

If

$$P^2 = P \qquad \text{and} \qquad PP^H = P^H P,$$

then

$$P = P^H.$$

Normal Projection Lemma

Lemma

If

$$P^2 = P \qquad \text{and} \qquad PP^H = P^H P,$$

then

$$P = P^H.$$

In words:

A normal projection cannot be oblique.

It must be a Hermitian orthogonal projection.

Proof: Measure the Obliqueness

Let

$$M = P - P^H P.$$

If $M = 0$, then

$$P = P^H P.$$

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$$M = P - P^H P.$$

If $M = 0$, then

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Taking Hermitian conjugates:

$$P^H = P^H P.$$

So $P = P^H$.

Goal: prove $M = 0$.

Proof: Positivity Kills the Obliqueness

Compute:

$$MM^H = (P - P^H P)(P^H - P^H P).$$

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$$MM^H = PP^H - PP^H P - P^H PP^H + P^H PP^H P.$$

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Compute:

$$MM^H = (P - P^H P)(P^H - P^H P).$$

Expand:

$$MM^H = PP^H - PP^H P - P^H PP^H + P^H PP^H P.$$

Using

$$P^2 = P, \qquad (P^H)^2 = P^H, \qquad PP^H = P^H P,$$

each term simplifies:

$$MM^H = PP^H - PP^H - PP^H + PP^H = 0.$$

By positivity, $MM^H = 0 \Rightarrow M = 0$.

Therefore Spectral Projections Are Hermitian

For a normal matrix A:

1. Spectral projections are polynomials: $P_i = f_i(A)$.
2. Polynomials of normal matrices are normal.
3. So each P_i is normal.
4. Each P_i is a projection: $P_i^2 = P_i$.
5. Normal projection lemma gives $P_i = P_i^H$.

Conclusion

For normal A, all spectral projections are Hermitian symmetric.

Hermitian Projection Means Orthogonal Projection

This is the same logic as the real case from the projection lecture:

$$P^2 = P, \quad P^T = P \quad \Longrightarrow \quad \mathrm{Col}(P) \perp \mathrm{Null}(P).$$

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Now we replace transpose by Hermitian conjugate:

$$P^2 = P, \quad P^H = P \quad \Longrightarrow \quad \mathrm{Col}(P) \perp_H \mathrm{Null}(P).$$

Same theorem, complex geometry.

Why? The Same Proof with H

General fact:

$$\mathrm{Col}(B^H) \perp_H \mathrm{Null}(B).$$

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Indeed, if $\mathbf{y} = B^H\mathbf{u}$ and $B\mathbf{x} = 0$, then

$$\mathbf{y}^H\mathbf{x} = (B^H\mathbf{u})^H\mathbf{x} = \mathbf{u}^H B\mathbf{x} = 0.$$

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Apply this to $B = P$ and use $P^H = P$:

$$\mathrm{Col}(P) = \mathrm{Col}(P^H) \perp_H \mathrm{Null}(P).$$

Apply This to Spectral Projections

For normal A, we proved:

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Apply This to Spectral Projections

For normal A, we proved:

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So each P_i is a Hermitian orthogonal projection:

$$\mathrm{Col}(P_i) \perp_H \mathrm{Null}(P_i).$$

But

$$\mathrm{Col}(P_i) = E_{\lambda_i}, \qquad \mathrm{Null}(P_i) = \bigoplus_{j \neq i} E_{\lambda_j}.$$

Therefore each eigenspace is perpendicular to all the others.

The Converse Direction

Suppose the spectral projections are Hermitian:

$$P_i^H = P_i.$$

Write the spectral decomposition:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

The Converse Direction

Suppose the spectral projections are Hermitian:

$$P_i^H = P_i.$$

Write the spectral decomposition:

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

Then

$$A^H = \overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k.$$

Because $P_i P_j = 0$ for $i \neq j$, both AA^H and $A^H A$ equal

$$\sum_i |\lambda_i|^2 P_i.$$

The Projection Criterion

We have proved:

$$\boxed{A \text{ normal} \quad \Longleftrightarrow \quad P_i^H = P_i \text{ for every spectral projection } P_i.}$$

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This is stronger than just diagonalizable.

It says:

The spectral decomposition is made of orthogonal projections.

Now the geometry becomes visible.

The Main Story Is Bigger

Hermitian symmetric, skew-Hermitian symmetric, and unitary matrices are all special cases of normal matrices.

But the main theorem is **not merely**:

$$\text{normal} \quad \Longrightarrow \quad \text{diagonalizable.}$$

That sentence loses the geometry.

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Hermitian symmetric, skew-Hermitian symmetric, and unitary matrices are all special cases of normal matrices.

But the main theorem is **not merely**:

$$\text{normal} \quad \Longrightarrow \quad \text{diagonalizable.}$$

That sentence loses the geometry.

The real statement has **two parts**:

$$\boxed{A \text{ normal} \quad \Longrightarrow \quad \begin{array}{c} \text{diagonalizable} \\ \text{and } P_i^H = P_i \text{ for every spectral projection } P_i \end{array}}$$

The Projection Version

From previous lectures:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k, \qquad P_i = f_i(A).$$

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$$\boxed{A \text{ normal} \quad \Longleftrightarrow \quad P_i^H = P_i \quad \text{for all } i.}$$

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$$\boxed{A \text{ normal} \quad \Longleftrightarrow \quad P_i^H = P_i \quad \text{for all } i.}$$

Meaning: every eigenspace projection is a Hermitian orthogonal projection.

The Geometric Version

Each spectral projection extracts one eigenspace component:

$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + \cdots + P_k\mathbf{v}.$$

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So the whole space decomposes as an **orthogonal direct sum**:

$$\boxed{\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.}$$

Key: every vector splits into orthogonal eigenvector components.

Unitary Diagonalization

How Do We Build the Diagonalizing Matrix?

We now know:

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} E_{\lambda_2} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

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To diagonalize A, we need an orthonormal basis inside each eigenspace.

But we already have the eigenspace projections:

$$P_i^H = P_i, \qquad \mathrm{Col}(P_i) = E_{\lambda_i}.$$

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But we already have the eigenspace projections:

$$P_i^H = P_i, \qquad \mathrm{Col}(P_i) = E_{\lambda_i}.$$

Question: How do we extract orthonormal basis vectors from P_i?

Use the Old Tool: Diagonal Cross-Filling

From the orthogonal projection lectures:

Previous Result

If $P^2 = P$ and $P^H = P$, then diagonal cross-filling of P produces orthonormal basis vectors for $\mathrm{Col}(P)$.

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Previous Result

If $P^2 = P$ and $P^H = P$, then diagonal cross-filling of P produces orthonormal basis vectors for $\mathrm{Col}(P)$.

In symbols:

$$P = \mathbf{u}_1\mathbf{u}_1^H + \cdots + \mathbf{u}_r\mathbf{u}_r^H,$$

where

$$\mathbf{u}_a^H\mathbf{u}_b = \delta_{ab}.$$

Same logic as before, with H replacing T.

Apply It to Every Spectral Projection

For each eigenvalue λ_i:

$$P_i = \mathbf{u}_{i,1}\mathbf{u}_{i,1}^H + \cdots + \mathbf{u}_{i,r_i}\mathbf{u}_{i,r_i}^H.$$

The vectors

$$\mathbf{u}_{i,1}, \ldots, \mathbf{u}_{i,r_i}$$

form an orthonormal basis of

$$E_{\lambda_i} = \mathrm{Col}(P_i).$$

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The vectors

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form an orthonormal basis of

$$E_{\lambda_i} = \mathrm{Col}(P_i).$$

Since different eigenspaces are already perpendicular, putting all these vectors together gives an orthonormal basis of $\mathbb{C}^n$.

Collect the Vectors into Ω

Collect all orthonormal eigenvectors as columns:

$$\Omega = \begin{pmatrix} | & | & & | \\ \mathbf{u}_{1,1} & \mathbf{u}_{1,2} & \cdots & \mathbf{u}_{k,r_k} \\ | & | & & | \end{pmatrix}.$$

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Because the columns are orthonormal:

$$\Omega^H \Omega = I.$$

So Ω is **unitary**, and

$$\Omega^{-1} = \Omega^H.$$

Unitary Diagonalization

Each column of Ω is an eigenvector of A.

So $A\Omega = \Omega\Lambda$, where

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{pmatrix}$$

with repetitions according to eigenspace dimensions

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with repetitions according to eigenspace dimensions

Multiply by Ω^H:

$$\boxed{\Omega^H A \Omega = \Lambda.}$$

Normal matrices are unitarily diagonalizable.

Example Matrix

Consider

$$A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

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It is real symmetric, so

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It is real symmetric, so

$$A^H = A^T = A.$$

Therefore A is normal.

The eigenvalues are

$$0, -3, -3.$$

Spectral Projections for the Example

The radical polynomial is

$$t(t+3).$$

The Lagrange spectral projections are:

$$P_0 = \frac{A+3I}{3} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and

$$P_{-3} = I - P_0 = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Diagonal Cross-Fill P_0

The projection

$$P_0 = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

has rank 1.

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has rank 1.

Diagonal cross-filling gives:

$$P_0 = \mathbf{u}_0 \mathbf{u}_0^H, \qquad \mathbf{u}_0 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

So $\mathbf{u}_0$ is an orthonormal basis for E_0.

Diagonal Cross-Fill P_{-3}

For the -3 eigenspace:

$$P_{-3} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Diagonal Cross-Fill P_{-3}

For the -3 eigenspace:

$$P_{-3} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Diagonal cross-filling gives one possible decomposition:

$$P_{-3} = \mathbf{u}_1\mathbf{u}_1^H + \mathbf{u}_2\mathbf{u}_2^H,$$

where

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad \mathbf{u}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

Stack the Rank-One Pieces

We have decomposed the two spectral projections:

$$\underbrace{P_0}_{E_0} = \mathbf{u}_0\mathbf{u}_0^H, \qquad \underbrace{P_{-3}}_{E_{-3}} = \mathbf{u}_1\mathbf{u}_1^H + \mathbf{u}_2\mathbf{u}_2^H.$$

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We have decomposed the two spectral projections:

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Since spectral projections add to the identity:

$$P_0 + P_{-3} = I.$$

Therefore

$$\mathbf{u}_0\mathbf{u}_0^H + \mathbf{u}_1\mathbf{u}_1^H + \mathbf{u}_2\mathbf{u}_2^H = I.$$

This Is $\Omega\Omega^H = I$

Stack the vectors as columns:

$$\Omega = \underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\text{colored eigenvectors}}.$$

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Then

$$\underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\Omega} \underbrace{\begin{pmatrix} \mathbf{u}_0^H \\ \mathbf{u}_1^H \\ \mathbf{u}_2^H \end{pmatrix}}_{\Omega^H} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I}.$$

So $\Omega\Omega^H = I$.

Reverse the Order

Now Ω is a square matrix.

From previous lectures:

Square Matrix Rule

If two square matrices satisfy $AB = I$, then also $BA = I$.

Reverse the Order

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From previous lectures:

Square Matrix Rule

If two square matrices satisfy $AB = I$, then also $BA = I$.

Apply this to

$$\Omega\Omega^H = I.$$

Reverse the order:

$$\Omega^H\Omega = I.$$

The reversed order is the inner-product table.

$\Omega^H\Omega = I$ — Inner Product Table

Spelled out:

$$\underbrace{\begin{pmatrix} \mathbf{u}_0^H \\ \mathbf{u}_1^H \\ \mathbf{u}_2^H \end{pmatrix}}_{\Omega^H} \underbrace{\begin{pmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 \end{pmatrix}}_{\Omega} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{inner product table}} .$$

$\Omega^H\Omega = I$ — Inner Product Table

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That means

$$\mathbf{u}_a^H \mathbf{u}_b = \delta_{ab}.$$

This proves the stacked eigenvectors are orthonormal.

Assemble the Unitary Matrix

Collect the orthonormal eigenvectors:

$$\Omega = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}.$$

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The columns are orthonormal, so

$$\Omega^H \Omega = I.$$

This is not an arbitrary eigenvector matrix. It is unitary.

The Diagonalization

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Therefore

$$\boxed{\Omega^H A \Omega = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}.}$$

Diagonal cross-filling of spectral projections produced the unitary diagonalization.

Eigenvalue Geometry

Return to the Special Cases

We defined three special kinds of normal matrices:

Type	**Matrix condition**
Hermitian symmetric	$A^H = A$
Skew-Hermitian symmetric	$A^H = -A$
Unitary	$A^H A = AA^H = I$

Return to the Special Cases

We defined three special kinds of normal matrices:

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Hermitian symmetric	$A^H = A$
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Unitary	$A^H A = AA^H = I$

Now normality lets us read these conditions from eigenvalues.

The matrix equation becomes a picture in the complex plane.

Start from Normal Spectral Decomposition

For a normal matrix,

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k, \qquad P_i^H = P_i.$$

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For a normal matrix,

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Taking Hermitian conjugate:

$$A^H = \overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k.$$

So A^H acts on the same orthogonal eigenspaces, but with conjugated eigenvalues.

Key: special matrix equations become restrictions on each λ_i.

Hermitian Symmetric Means Real Eigenvalues

Suppose

$$A^H = A.$$

Compare the two spectral decompositions:

$$\overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

Hermitian Symmetric Means Real Eigenvalues

Suppose

$$A^H = A.$$

Compare the two spectral decompositions:

$$\overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k = \lambda_1 P_1 + \cdots + \lambda_k P_k.$$

Therefore, for every eigenvalue:

$$\overline{\lambda_i} = \lambda_i.$$

Conclusion

Hermitian symmetric normal matrices have eigenvalues on the **real axis**.

Skew-Hermitian Means Imaginary Eigenvalues

Suppose

$$A^H = -A.$$

Compare:

$$\overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k = -\lambda_1 P_1 - \cdots - \lambda_k P_k.$$

Skew-Hermitian Means Imaginary Eigenvalues

Suppose

$$A^H = -A.$$

Compare:

$$\overline{\lambda_1} P_1 + \cdots + \overline{\lambda_k} P_k = -\lambda_1 P_1 - \cdots - \lambda_k P_k.$$

Therefore, for every eigenvalue:

$$\overline{\lambda_i} = -\lambda_i.$$

Conclusion

Skew-Hermitian normal matrices have eigenvalues on the **imaginary axis**.

Unitary Means Unit-Circle Eigenvalues

Suppose

$$U^H U = I.$$

On the λ_i-eigenspace:

$$U \text{ acts by } \lambda_i, \qquad U^H \text{ acts by } \overline{\lambda_i}.$$

Unitary Means Unit-Circle Eigenvalues

Suppose

$$U^H U = I.$$

On the λ_i-eigenspace:

$$U \text{ acts by } \lambda_i, \qquad U^H \text{ acts by } \overline{\lambda_i}.$$

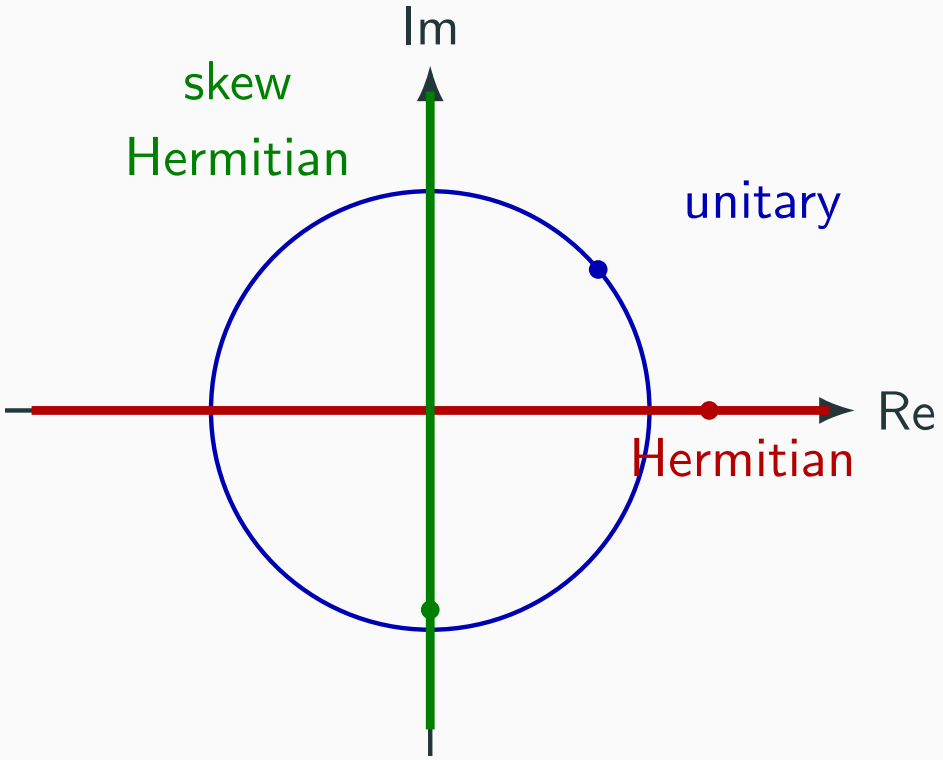
So $U^H U = I$ gives

$$\overline{\lambda_i}\lambda_i = 1.$$

Conclusion

Unitary normal matrices have eigenvalues on the **unit circle**: $|\lambda_i| = 1$.

The Complex-Plane Picture



$A^H = A$	eigenvalues on real axis
$A^H = -A$	eigenvalues on imaginary axis
$U^H U = I$	eigenvalues on unit circle

What Survives Over Real Numbers?

The true machine is complex:

$$A \text{ normal} \quad \Longrightarrow \quad \Omega^H A \Omega = \Lambda.$$

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For real applications, we ask:

When can the whole computation stay real?

That means:

- eigenvalues are real,
- spectral projections have real entries,
- diagonal cross-filling uses real vectors.

Real Symmetric Matrices Stay Real

If A is real symmetric:

$$A^T = A.$$

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Therefore all eigenvalues are real.

The Lagrange projection formula uses real roots:

$$P_i = f_i(A), \qquad f_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

Since A and the λ_i are real, every P_i is real.

Real Orthogonal Diagonalization

For a real symmetric matrix:

$$P_i^H = P_i \quad \text{becomes} \quad P_i^T = P_i.$$

Real Orthogonal Diagonalization

For a real symmetric matrix:

$$P_i^H = P_i \quad \text{becomes} \quad P_i^T = P_i.$$

Diagonal cross-filling uses only real numbers and produces real orthonormal eigenvectors.

Stack them into a real matrix Q:

$$Q^T Q = I, \qquad Q^T A Q = \Lambda.$$

Real Specialization

Real symmetric matrices admit real orthogonal diagonalization.

Why Skew-Symmetric Real Matrices Leave $\mathbb{R}$

If A is real skew-symmetric:

$$A^T = -A.$$

Why Skew-Symmetric Real Matrices Leave $\mathbb{R}$

If A is real skew-symmetric:

$$A^T = -A.$$

Then $A^H = A^T = -A$, so A is skew-Hermitian.

Skew-Hermitian eigenvalues lie on the imaginary axis:

$$\lambda \in i\mathbb{R}.$$

Why Skew-Symmetric Real Matrices Leave $\mathbb{R}$

If A is real skew-symmetric:

$$A^T = -A.$$

Then $A^H = A^T = -A$, so A is skew-Hermitian.

Skew-Hermitian eigenvalues lie on the imaginary axis:

$$\lambda \in i\mathbb{R}.$$

So nonzero eigenvalues are usually not real.

This is why the complex world is necessary.

Real-World Summary

Real matrix type	Complex view	Real outcome
$A^T = A$	Hermitian	real orthogonal diagonalization
$A^T = -A$	skew-Hermitian	imaginary eigenvalues
general real A	may be non-normal	no orthogonal eigenspace guarantee

Real-World Summary

Real matrix type	Complex view	Real outcome
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$A^T = -A$	skew-Hermitian	imaginary eigenvalues
general real A	may be non-normal	no orthogonal eigenspace guarantee

Hermitian geometry is the real god: real symmetric theory is its real shadow.

Final Picture of Normal Matrices

Normal matrices are the matrices whose spectral decomposition is orthogonal:

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

Final Picture of Normal Matrices

Normal matrices are the matrices whose spectral decomposition is orthogonal:

$$\mathbb{C}^n = E_{\lambda_1} \overset{\perp_H}{\oplus} \cdots \overset{\perp_H}{\oplus} E_{\lambda_k}.$$

Special normal matrices add restrictions on eigenvalue locations:

Hermitian	real axis
Skew-Hermitian	imaginary axis
Unitary	unit circle

Key: orthogonal eigenspaces first; eigenvalue geometry second.